📘 MODULE 9: TIME IN SPLUNK

9.1 Concept

Every event has _time, and **time is the first filter Splunk uses to decide which data to read**. A narrow time range is the single easiest performance win.

9.2 Time range

- **Time picker** (right of the search bar): Last 15 minutes, 24 hours, 7 days, Custom, All time.
- **In the search**, earliest and latest **override** the picker.

spl

index=main earliest=-24h latest=now

index=main earliest=-7d

index=main earliest=-15m

index=main earliest="10/07/2026:00:00:00" latest="10/08/2026:00:00:00"

9.3 Relative time syntax

Piece	Meaning
-15m	15 minutes ago
-24h	24 hours ago
-7d	7 days ago
-1mon	1 month ago (units: s m h d w mon y)
now	Current time
@d	**Snap** to the start of the day
-1d@d	Start of yesterday
-7d@w1	Start of the week (Monday)

Yesterday only: earliest=-1d@d latest=@d.

9.4 Special time fields

- _time: the event time (stored as **epoch**, seconds since 1970).
- _indextime: when Splunk indexed it. A large gap between _time and _indextime means delayed ingestion or a wrong timestamp.

- Auto-extracted: date_hour, date_minute, date_wday (may be absent or unreliable on some data).

9.5 Worked queries

Format time

spl

index=_internal | head 5 | eval readable=strftime(_time,"%Y-%m-%d %H:%M:%S")

| table _time readable

What it does: converts epoch to readable text. Format codes: %Y year, %m month, %d day, %H hour (24h), %M minute, %S second, %F=%Y-%m-%d, %T=%H:%M:%S, %A weekday name.

Events per hour

spl

index=_internal sourcetype=splunkd earliest=-24h

| timechart span=1h count by log_level

What it does: counts events in 1-hour buckets, one column per log_level.
Line-by-line: timechart always uses _time for the x-axis. span=1h is the bucket size. by log_level splits into series.
Expected result: a table with _time and columns per level. Click **Visualization** → Line/Column chart.
Common mistakes: a too-small span creates thousands of points, so use limit or a bigger span. With no events in a bucket, timechart still shows 0.

bin with stats

spl

index=_internal sourcetype=splunkd_access earliest=-24h

| bin _time span=1h

| stats count by _time, status

What it does: bin rounds each _time down to the hour so stats can group by it. timechart is the shortcut. Use bin+stats when you need custom groupings.

Time since an event

spl

... | stats latest(_time) AS last_seen by host

| eval minutes_silent=round((now()-last_seen)/60)

| where minutes_silent>60

SOC use: finds hosts that stopped sending logs, which is a classic “silent host” detection.

9.6 Performance

- Splunk stores data in time-ordered buckets. A narrow range means **fewer buckets opened**.
- earliest=-15m on a busy index is far faster than All time.
- Use **real-time** searches sparingly because they hold resources continuously.
- Combine: **specific index + sourcetype + narrow time** in the base search.

9.7 Hands-on

1. Show splunkd ERROR events from the last 15 minutes.
2. Show events only from yesterday.
3. Make a timechart of events per 10 minutes for the last 4 hours.
4. For each host, find the last time it sent data and how many minutes ago that was.

9.8 Troubleshooting

Symptom	Cause
No results but data exists	Time range too narrow, or events have a wrong or future timestamp
Events appear hours off	Time zone mismatch or wrong TIME_FORMAT in props (Module 23)
earliest in search ignored	Check spelling, and quoting for absolute dates
Timechart shows odd gaps	Span too small for the data rate

9.9 Interview Q&A

- **How does time affect performance?** Splunk reads only buckets that overlap the range.
- **earliest/latest vs the picker?** The search’s values override the picker.

- **_time vs _indextime?** Event occurrence vs ingestion time.
- **timechart vs chart?** timechart is always x-axis time. chart can use any field.
- **What does @d mean?** Snap to the beginning of the day.

9.10 Revision

earliest/latest override the picker · -15m -24h -7d · @d snaps · _time is epoch · narrow time = faster.